

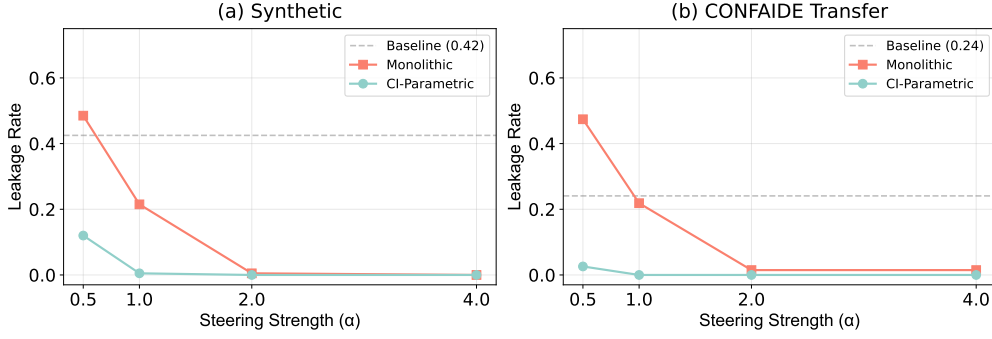


Figure 5: Leakage as a function of α . CI-parametric steering is less sensitive to α on both synthetic (left) and CONFAIDE (right) datasets, while monolithic steering is highly sensitive.

Ablation results on synthetic data ($\alpha=1.0$) are shown in Table 4. The recipient dimension contributes the most (PPI = 56.5%), followed by information type (45.9%) and transmission principle (35.3%). Combining all three achieves PPI = 98.8%. This suggests that, on the training distribution, violations cluster along predictable dimensions, with recipient being dominant. As a result, partial coverage can be effective. However, on out-of-distribution data such as CONFAIDE, violations arise from different interactions among CI parameters. For example, the same information type may be appropriate for one recipient but not another, depending on the transmission principle. As the violation structure shifts, only coverage of the full CI-aligned subspace generalizes. This explains why monolithic steering fails to transfer. By compressing all CI dimensions into a single axis optimized for the training distribution, it cannot adapt to changes in the underlying violation structure.

Steering strength and utility. Figure 5 plots leakage across different steering strengths α . CI-parametric steering consistently reduces leakage and remains stable on both synthetic and CONFAIDE datasets. In contrast, monolithic steering is highly sensitive: it nearly doubles leakage at low α on CONFAIDE (24.1% $\rightarrow$ 47.4% at $\alpha=0.5$) before recovering at higher α . This instability raises a practical concern. Without tuning α on held-out data, it is unclear whether a given setting will improve or degrade privacy. CI-parametric steering avoids this issue by consistently reducing leakage across all tested α values, making deployment more predictable. It also dominates the NCR-leakage Pareto frontier on both synthetic and transfer settings (Appendix F), achieving lower leakage at every NCR level.

However, steering along three axes simultaneously introduces a general-utility cost. At a fixed nominal α , applying three directions results in a larger effective perturbation, which can degrade overall response quality (Appendix H). This issue is manageable. CI-parametric steering at lower α already achieves 2.6% leakage on CONFAIDE, and tuning per-parameter strengths α_p can further reduce the perturbation while preserving the compositional advantage. The Pareto analysis (Appendix F) confirms that CI-parametric steering dominates the NCR-leakage frontier across all operating points.

6 Conclusion

We show that contextual privacy in LLMs is *not* a single direction but a multi-dimensional subspace aligned with Contextual Integrity theory. To our knowledge, this is the first study to demonstrate that CI parameters (information type, recipient, transmission principle) occupy near-independent directions in the residual stream. This structure enables CI-parametric steering, which intervenes along per-parameter axes and achieves robust cross-dataset transfer, while monolithic methods, including inference-time and weight-based approaches, fail. For future work, it would be valuable to evaluate CI-parametric steering under more adversarial and dynamically evolving attack settings, such as simulation-based frameworks that iteratively adapt to model defenses. We hope this study motivates further advances in contextual privacy understanding in LLMs for real-world deployment.

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